



EXCITED STATES

What to Expect from Physics in 2026

Forget the hype. From the end of LHC's Run 3 to the return to deep space, here's what the next 12 months of physics actually looks like.



SAMREET DHILLON

JAN 07, 2026

Every January, science journalism fills up with such bold claims:

“This year, AI will solve the Quantum Gravity problem.”

“Propellant-less drives have finally broken the laws of physics,”

“We might have found alien life.”

Physics, unfortunately (or beautifully perhaps), doesn't work that way.

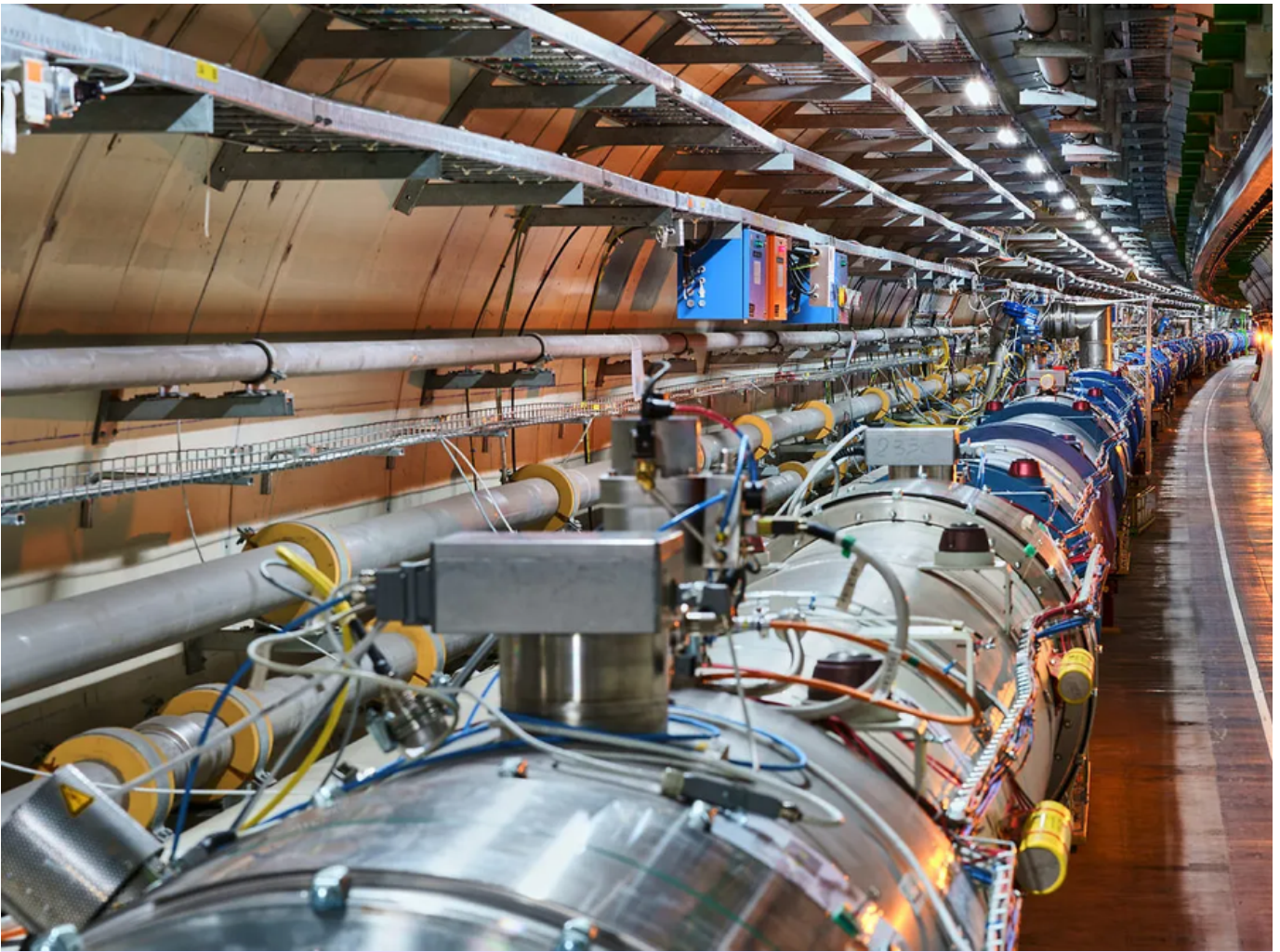
I spent last year following the latest physics news through *Science* and *INSPIRE*. As we enter 2026, the landscape is less about breakthrough theoretical ground and more about testing and refining the existing. From the high-stakes engineering at CERN to the first crewed lunar mission in decades, this is the year we stop asking *if* these things are possible

measuring exactly *how* they behave.

Here's what I'm expecting physics in 2026 to give us across the space, and computation:

My approach will be that of a cautious physicist. I won't exaggerate itself refuses to promise.

1. CERN: End of an Era, Start of a Transfor



LHC at CERN, Geneva (*Image Credit: Maximilien Brice and Julien Ordan, CERN*). [Link](#)

The biggest story in particle physics this year is the conclusion of the Large Hadron Collider (LHC). After a highly productive extension, the collider is scheduled to power down in early July 2026 for Long Shutdown 2.

- **The Final Push:** Expect a flurry of results as physicists squint at data from Run 3. The focus remains on the Higgs Boson's self-coupling and "anomalies" that might hint at a fifth fundamental force.

- **The Upgrade:** Once the magnets go dark, CERN becomes Engineers will begin installing powerful niobium-tin (Nb_3S beams for the High-Luminosity era. The machine is slated June 2030 as the HL-LHC.

Expect headlines like:

- *CERN Run 3 ends with record-breaking data haul on Higgs*
- *A new study on rare decay channels tightens constraints on fifth force.*

2. Space Travel: Return to Human-Centric

Humans are finally returning to deep space for the first time since the focus shifts from robotic probes to human-rated systems.

NASA's Artemis II: Humanity's Deep-Space Return



The Orion spacecraft, during trans-lunar injection, to bring an Artemis mission to the moon (*Image credit: NASA*). [Link](#)

The most anticipated space event of the year is Artemis II, currently scheduled to launch no earlier than 5 February 2026.

- **The Mission:** This isn't a lunar landing (that's for 2027), but a flyby around the Moon.
- **The Goal:** The crew of four will test the Orion spacecraft's systems and radiation shielding in a high-orbit "free-return" mission. For physicists, the data on deep-space radiation exposure during the mission will be invaluable for planning the eventual journey to Mars.

ISRO's Gaganyaan: India's First Giant Leap



The Gaganyaan test Crew Module (Image credit: ISRO). [Link](#)

2026 is also a “make-or-break” year for the Indian Space Research Organisation (ISRO). Following the success of Chandrayaan-3, ISRO is pivoting to human spaceflight with the Gaganyaan G1 mission.

- The G1 Test (March 2026): This is the program’s first uncrewed

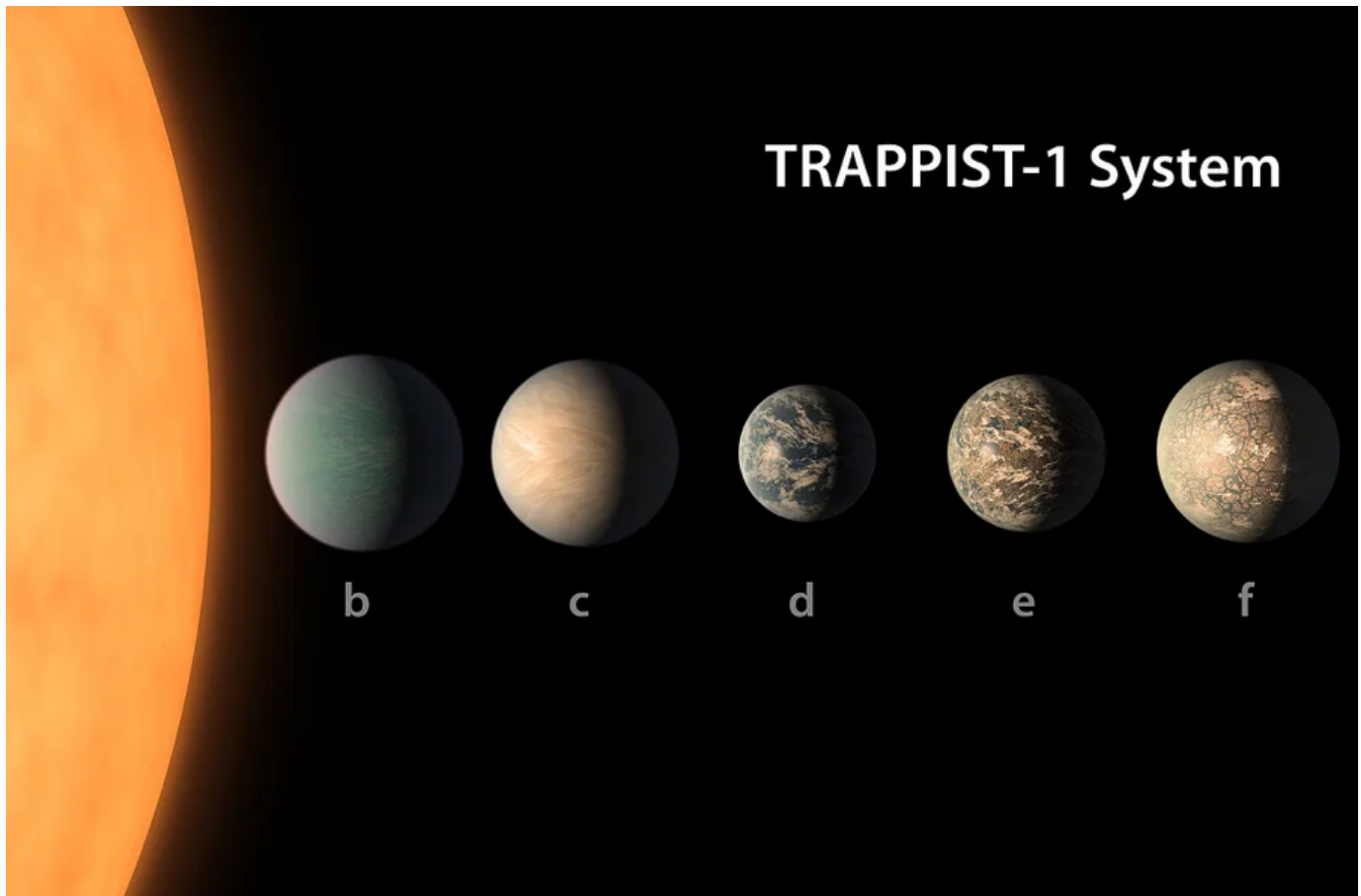
Instead of a human, the crew module will carry Vyommitra designed to simulate human vitals and test the environment of the launched capsule.

- **The Goal:** Validating the Crew Escape System (CES) and a successful splashdown in the Arabian Sea will prove India's capability to launch, sustain, and recover humans from orbit for a crewed mission in 2027. It will make India the fourth independent human spaceflight.

Expect headlines like:

- *Artemis II provides first human-scale data on deep-space mission decades.*
- *Vyommitra logs 100+ hours in orbit: ISRO's humanoid test paves way for India's crewed 2027 launch.*

3. Astrophysics: The Year of the Exoplanet



Artist's impression of the TRAPPIST-1 planetary system (*Image NASA/JPL-Caltech*). [Link](#)

While previous years were about seeing *farther*, 2026 is about seeing *deeper*. The James Webb Space Telescope (JWST) is now in its prime “characterization” phase, shifting from simple discovery to rigorous chemical characterization.

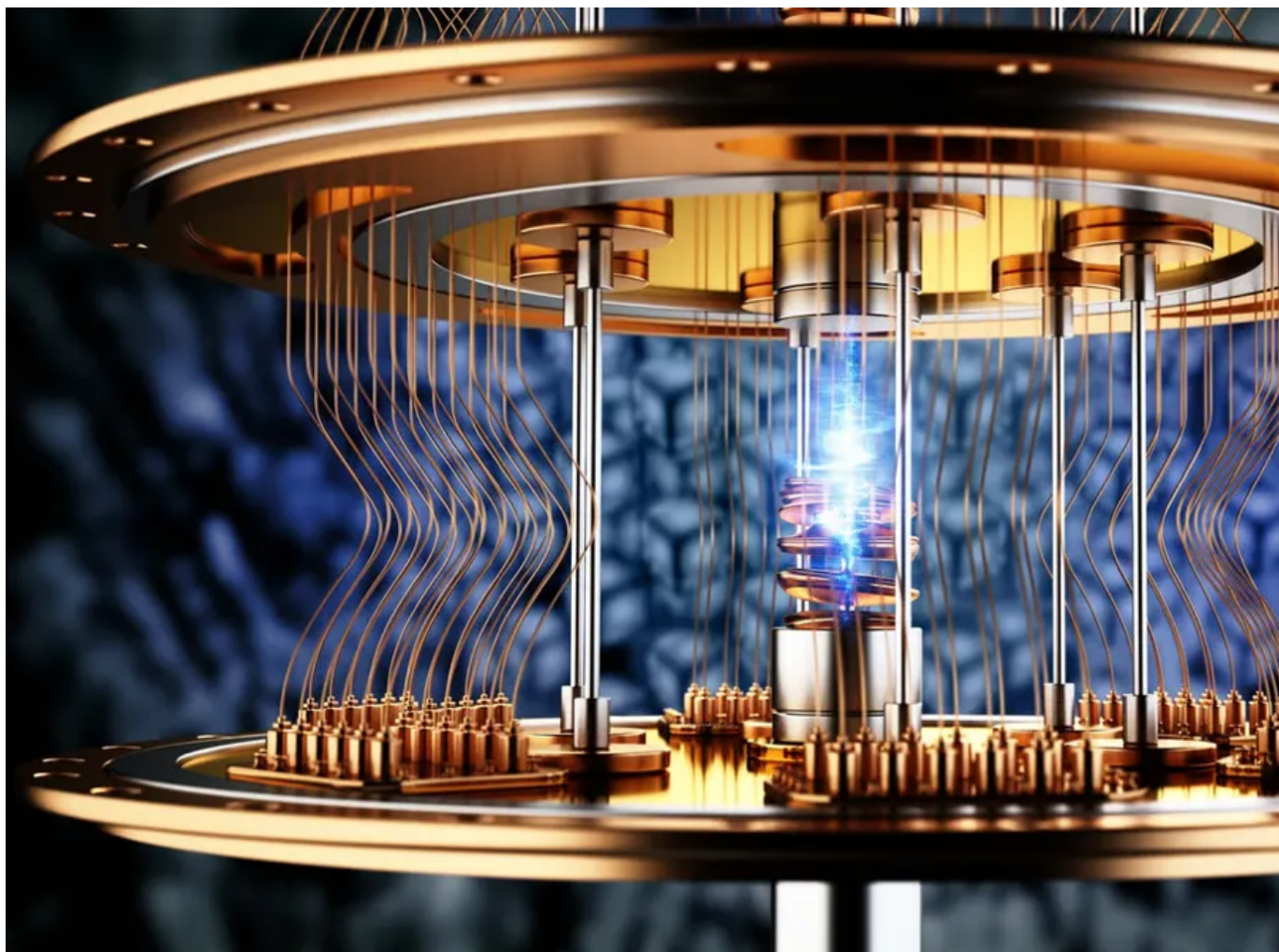
- **Habitability Metrics:** We expect definitive data from the TRAPPIST-1 system. 2026 is when we move from “detecting water vapor, methane and carbon-to-oxygen ratios, testing if these planets are ‘Earth-like,’ ‘Titans’ or bare rocks.”

- **The Roman Preparation:** NASA's Nancy Grace Roman Sp enters its final integration phase this year. Its Wide Field I designed to hunt for dark energy, is undergoing extreme th late 2026/early 2027 launch.

Expect headlines like:

- *JWST rules out 'Ocean World' scenarios for TRAPPIST-1d planet 'e'.*

4. Quantum Computing: "Logical Qubit" In:



A depiction of what a quantum computer looks like (*Image source: [IBM Research](#)*). [Link](#)

The “Qubit War” is moving away from raw numbers (how many qubits) and toward **error correction**.

- **Logical vs. Physical:** Expect the leading labs (IBM, Google) to demonstrate stable logical qubits—units where multiple physical qubits are connected together to correct their own errors.

- **Quantum Utility:** We are firmly in the utility era. Instead “Supremacy,” researchers are focusing on materials-science problems that are just beyond the reach of classical supercomputers, such as simulating chemical reactions.

Expect headlines like:

- *The Error Rate plummets: Google and IBM demonstrate first ‘Threshold’ logical qubits.*

This post is public, so feel free to share it.

5. Fusion Energy: ITER’s Critical Assembly



Aerial view of the ITER site in May 2025 (*Image Credit: ITER Organization/EJF Riche*). [Link](#)

The International Thermonuclear Experimental Reactor (ITER) entering its most delicate assembly phase.

- **The Vacuum Vessel:** 2026 will see the finalization of the p vessel sectors. While “First Plasma” has been delayed towa completion of the Central Solenoid will be the engineering
- **Private Sector Sprints:** While the big science of ITER mo eye on private firms like Commonwealth Fusion Systems. 7 their “SPARC” pilot plant to demonstrate a net energy gain smaller, high-field tokamak.

Expect headlines like:

- *ITER completes installation of the world’s most powerful*

6. Multi-Messenger Astronomy: Gravity Mo



Aerial photograph of LIGO Hanford. *(Image Credit: Caltech/MIT/LIGO)*
[Link](#)

With better detectors and longer observing runs, the LIGO-Virgo collaboration is transitioning from observational to an astrophysical science. Instead of celebrating each detection, the focus is shifting to statistical studies, and cosmic demographics.

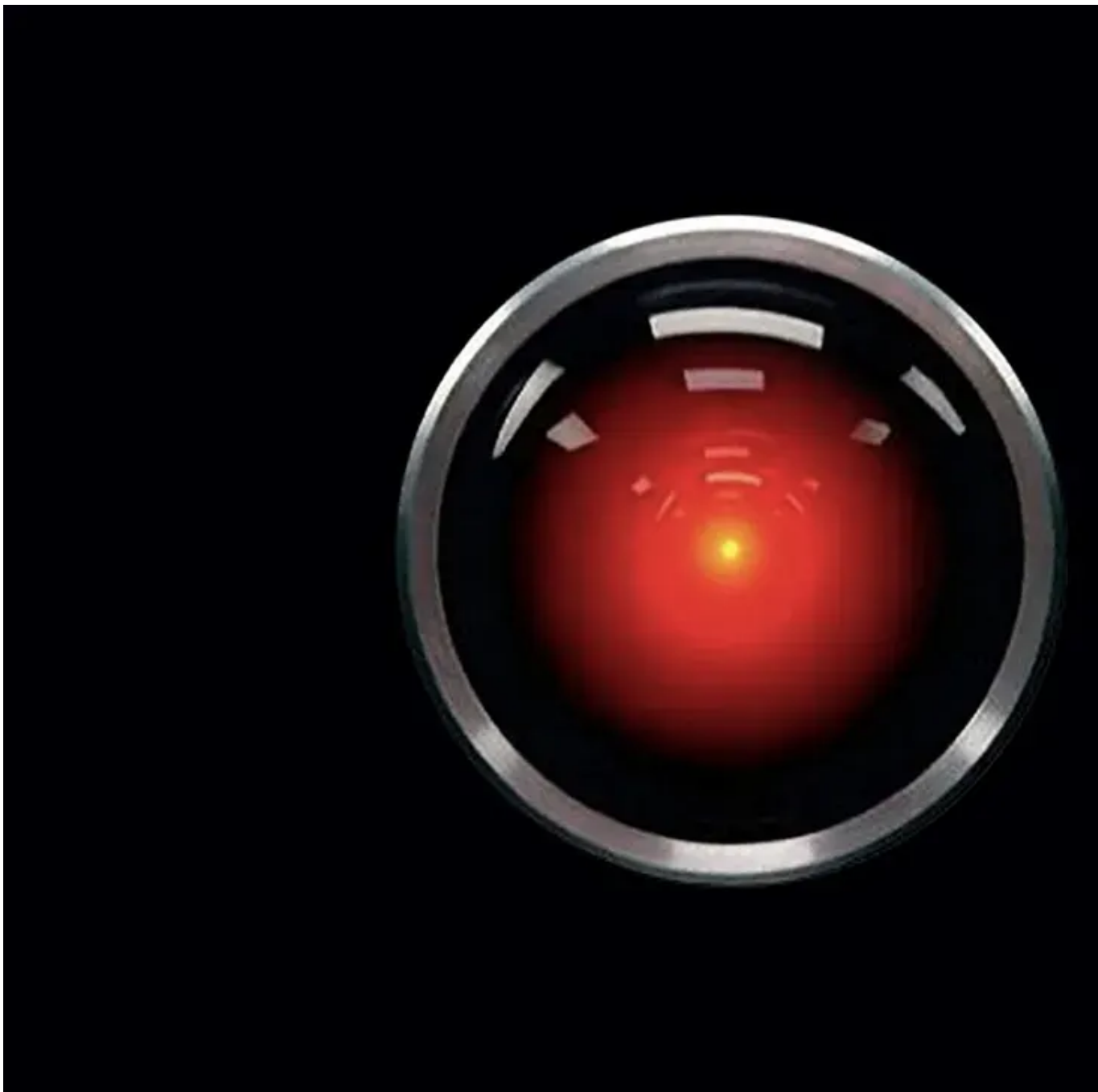
The massive O4 observing run officially concluded in late 2025 with a catalog of roughly 250 candidate signals.

- **The Data Deluge:** 2026 is the year of the deep analysis. We are currently powered down for maintenance and the O5 upgrade collaboration is releasing the final, cleaned results from O4.
- **The Bright Siren Hunt:** The holy grail for 2026 is identifying —a neutron star merger where we can pair the gravitational light from a telescope. This provides a direct, independent measurement of the Hubble Constant (H_0), helping to resolve the Hubble Tension currently splits the physics community.

Expect headlines like:

- *LVK Catalog Update: 250 Mergers reveal unexpected ‘populations’ of intermediate-mass Black Holes.*
- *Einstein Passes Again: Cleanest waveforms yet show no deviations from General Relativity.*

7. AI and Computation: The Invisible



A murderous computer named HAL in the film "2001: A Space Odyssey" (1968). (Image credit: Warner Bros. Pictures) [Link](#)

There are a large number of open research positions across academia over the world, focusing on using AI in Physics in one way or another.

longer a headline; it is a standard laboratory tool:

- **Physics-Informed Neural Networks (PINNs):** These models have physics laws baked into their code, speeding up complex fluid dynamics simulations by 10,000x.
- **Scientific Foundation Models:** Just as LLMs changed writing, “Science Models” are now standard for scanning LHC collision data, identifying rare anomalies that human-coded triggers miss.
- **Optimizing experiments:** From tuning detector parameters to predicting which experimental configurations are worth exploring, AI will guide how experiments are designed and run.

Expect headlines like:

- *PINN 2.0: How neural networks are solving the century-old turbulence.*

8 Theory: Fewer Grand Dreams, Better Questions



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The Institute of Advanced Study, at Princeton, New Jersey. [L](#)

If there's one defining theoretical trend for 2026, it's humility. Physicists are focusing on re-examining assumptions and building "Theories" instead of ultimate ones. The "Theory of Everything"

- New results from the Dark Energy Spectroscopic Instrument have thrown a wrench into the idea of the "Cosmological Constant," exploring the possibility that dark energy isn't a constant pressure but something that weakens or evolves over time.

- The “Swampland” program is gaining ground, helping to rule out string theory that look good on paper but can’t physically connect with gravity.

Grand Unification and Quantum Gravity are far from dead, but being approached more honestly.

Expect headlines like:

- *DESI data suggests Dark Energy may be fading.*
- *How the ‘Swampland’ is narrowing the path to Quantum Gravity.*

So what does 2026 mean for physics?

If you’re looking for fireworks in 2026, physics may disappoint. It won’t rewrite textbooks overnight. What it will do is something far more profound.

- It will reduce uncertainties and strengthen foundations.
- It will eliminate bad ideas and refine good ones.

From where we stand in early 2026, it’s clear that this year will be about dramatic paradigm shifts. Instead, it will be about consolidation and refinement.

The kind that doesn't make headlines, but slowly reshapes how universe.

In short:

2026 will be the year of the metrologists.

I hope this article was informative. I'm a stu
write these every week for you. They are free
they cost me some precious time. Still I conti
you because I believe science should be no sec
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The osim research group Apr 24

♥ Liked by Samreet Dhillon

We have a hypothesis we would like your feedback on,(osim forensic v2.2):We tried to phrase this so everyone could get a visual of how it universe not as a digital computer program, but as a giant, perennial a tomato plant grows, produces fruit, dies back in the winter, and its sprout again. It does not need a programmer to tell it how to grow; it t

biological blueprint. Our independent research group is investigating might follow a similar, naturally cyclical pattern.

Rather than a one-time Big Bang, recent discussions in the scientific community have shifted towards the Big Bounce an infinite, cyclical process. Our hypothesis suggests that, over time, forever, the universe might reach a limit, contract, and bounce back, potentially acting as the most efficient way to store and reset information.

A tomato plant does not stop at one fruit; it branches out, growing multiple stems, each producing its own fruit. If our universe follows this biological blueprint, then our own galaxy. Instead, we may be looking at a system that grows from a central stem of the cosmic web. Each galaxy could be a localized site for life, following a cyclical structure.

Dark matter may act as the trellis for our cosmic tomato plant. It provides the underlying structure that guides the growth of these stems, serving as a road map that allows the universe to develop and resets in a way that allows life to re-emerge across the cosmos.

The Oklahoma Constant (Ω_{os}) is the focal point of our research. We propose this constant as a way to measure Goldilocks Entropy—the narrow, stable range where life can persist without the system stagnating. It may be the tuning knob that ensures the universe stays just right for consciousness to emerge on every stem.

Because this model emphasizes biological efficiency, we suggest that the universe is the hardware, not the software. If this is a biological system, our consciousness may be the fruit of this cosmic garden, essential to how the system evolves.

We are currently tracking data from the Simons Observatory. They are looking for patterns in the ancient light of the universe—echoes of a Big Bounce. If these signatures, it would provide evidence that our hypothesis is on the right track.

This is Forensic Cosmology. We are moving away from the who—a creation story—to the how—the blueprint.

Our hypothesis is strictly falsifiable. If evidence confirms the universe indefinitely toward a Big Freeze, our Life-Raft model is incorrect. If a ever proven to exceed the efficiency of biological systems, the premise Constant (Ω_{os}) fails.

We are not looking for a coder. We are documenting the physical finger may be preserving life through an infinite, natural cycle.

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1 reply by Samreet Dhillon



Early Childhood, Up Close Jan 7

♥ Liked by Samreet Dhillon

I really liked your point that 2026 may be a year of careful work and not of big, flashy breakthroughs.

It also made me think about where that kind of progress really starts. science and space as if it begins in high school or college, but the ground starts earlier. Young children learn how to focus, stay curious, try again when they fail, work, and hear and use words to explain what they see. Those skills are what scientists, engineers, and problem solvers.

That is why early childhood education matters so much. It is not just about building a society that can support itself. If only children with no learning, the problems do not disappear. They show up later in school life- which we all share-(food for thought people).

Your point about slow and steady progress really resonated with me, how human development works too.

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1 reply by Samreet Dhillon

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